

OOP in Java

BCA — Semester III — 2025 — Answer Sheet
Subject Code: CACS204

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. What is command-line arguments? Write a program to accept two numbers and display their sum using command-line arguments.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Write a program to demonstrate exception handling of `ArrayIndexOutOfBoundsException`.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is string immutability? Explain with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Write a program to create multiple threads and execute them concurrently.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain the `Serializable` interface in java with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is multiple inheritance? Explain how multiple inheritance is implemented in Java?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is JDBC? Explain its architecture.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Explain nested and inner classes in java with examples. Discuss the scenarios where inner classes are useful.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Create a class Person (with attributes name and age, and a method display()). Derive a class Employee from Person with additional attributes employeeid and salary. Use constructors and override the display() method to include all details.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Build a form with Passenger Name, Flight Number, Seat Preference (Window, Aisle, Middle), and a checkbox for "I agree to baggage rules." Clicking Confirm Booking should save data to "flight.txt" if rules are accepted otherwise display "Accept baggage rules first."

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.